SPRINT 7 REVIEW REPORT

14.4.2026 - 28.4.2026

Sprint Goal

The goal of this sprint was to plan and carry out User Acceptance Testing and to perform a heuristic evaluation on the application to make sure the software meets the set requirements and criteria.

Completed Tasks

The following tasks were completed in this sprint:

- **Code clean up**
  - Duplication reduction: Some new classes were created to group together methods that are used by multiple classes.
  - Fixing best practices issues: “Final” statement was added to classes, attributes and functions that might need it. Code lines were split on to multiple rows.

- **SonarQube + Jenkins:**
  - SonarQube analysis stage was added to Jenkins pipeline
  - Instructions for using the SonarQube through Jenkins were added to the README file.

- **User Acceptance Testing (UAT)**
  - Test cases created based on the acceptance criteria
  - All tests carried out by four testers
  - Report for the results

- **Heuristic Evaluation**
  - Application analysed by four evaluators
  - Issues recorded and rated
  - Report for the evaluations

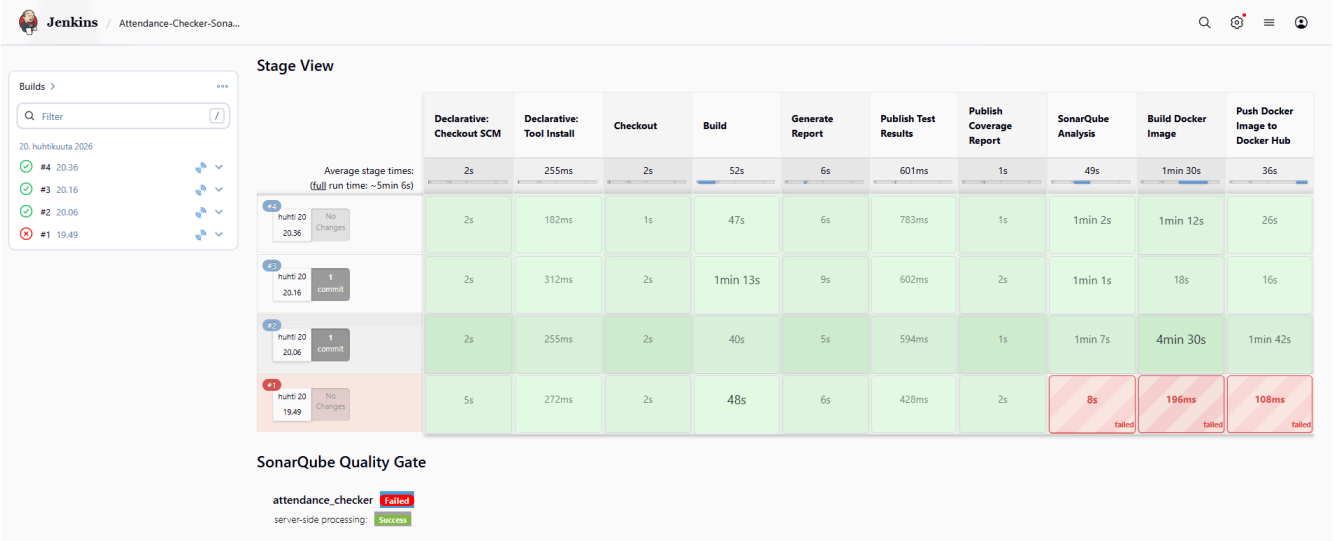
- **A sheet for recording bugs**
  - A Google Sheets file was created to keep track of bugs that might appear while testing and evaluating the software.

- **Package diagram**
  - A package diagram was created for the software containing all the major packages.

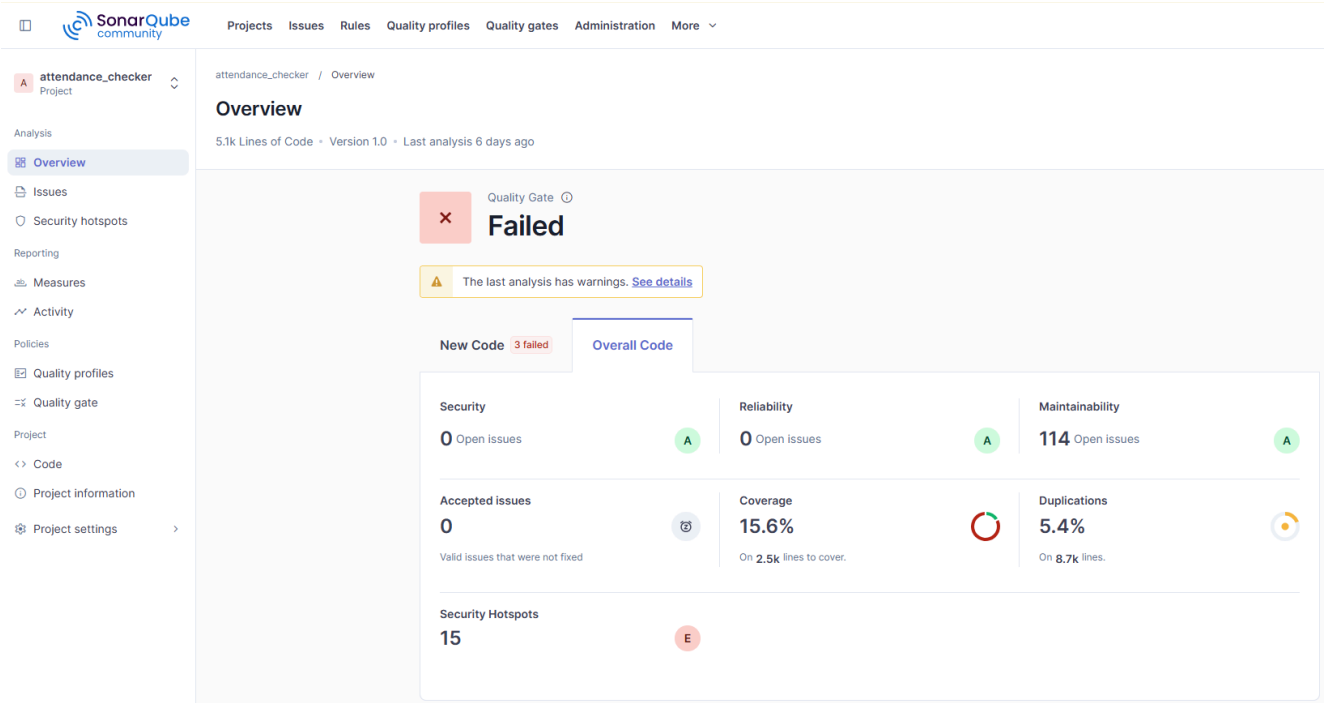
No serious challenges were encountered when these tasks were carried out.

SonarQube and Jenkins pipeline

SonarQube analysis stage was added to the Jenkins pipeline so whenever the project is built through Jenkins, a SonarQube analysis is performed on the code. The analysis can be viewed from the local machine. Here are screenshots of the Jenkins pipeline showing the SonarQube analysis stage, and of the SonarQube analysis resulting from the Jenkins pipeline. Note that the analysis was done before the final clone clean up so there are still some Security Hotspot issues shown in the screenshot.



Picture 1. Jenkins pipeline showing the SonarQube analysis stage.



Picture 2. The SonarQube analysis resulting from Jenkins pipeline.

Technical Changes

No major changes to the specifications of the software were applied. The software still contains the same plugins and dependencies, though SonarQube analysis was added to the Jenkins pipeline.

Code architecture was updated and modified to decrease duplicate code. Algorithms for calculating attendance percentages were moved to a single class in the utils-package. Views implement the newly created UIView interface. The UIComponent class was created to contain the top and side panel components of the views. The class creates them and returns the components to the views.

User Acceptance Testing (UAT) summary

All ten created test cases were successfully passed by all the testers. This means that the application has passed the User Acceptance Testing with 100% passing rate. The table below shows all the test cases results from the individual testers.

Test Case	Riikka K	Artem P	Jere P	Zachris Z
TC_LOGIN_01	Pass	Pass	Pass	Pass
TC_LOGIN_02	Pass	Pass	Pass	Pass
TC_STUDENTS_01	Pass	Pass	Pass	Pass
TC_CHECKS_01	Pass	Pass	Pass	Pass
TC_CHECKS_02	Pass	Pass	Pass	Pass
TC_ATTENDANCE_01	Pass	Pass	Pass	Pass
TC_ATTENDANCE_02	Pass	Pass	Pass	Pass
TC_REPORT_01	Pass	Pass	Pass	Pass
TC_REPORT_02	Pass	Pass	Pass	Pass
TC_LOGOUT_01	Pass	Pass	Pass	Pass

For a more detailed overview of the User Acceptance Testing and the test cases, refer to the User Acceptance Testing report:
https://github.com/ArtemPoliakov51/OTP1/blob/main/Documents/product_documents/attendance_checker_user_acceptance_testing_report.pdf

Heuristic evaluation summary

The average severity scores for each heuristic from all the evaluators, and total average of all the average scores:

No.	Heuristic	Riikka K	Artem P	Jere P	Zachris Z	Average total
1	H1-1: Simple & natural dialog	1.25	1	1	1	1.06
2	H1-2: Speak the users' language	1.5	1	1	1	1.13
3	H1-3: Minimize users' memory load	1	2	1	0	1
4	H1-4: Consistency	0.5	1	0	0	0.38
5	H1-5: Feedback	2.33	2	0	0	1.08
6	H1-6: Clearly marked exits	1.5	1	2	2	1.63
7	H1-7: Shortcuts	1	1	3	3	2
8	H1-8: Precise & constructive error messages	2.33	3	0	0	1.33
9	H1-9: Prevent errors	1.5	2	2	2	1.88
10	H1-10: Help and documentation	2	1	0	0	0.75

The most obvious errors or usability issues were the lack of keyboard navigation, and mission confirmation dialogs (error prevention). Especially in terms of accessibility, keyboard navigation is important.

This is a summary of the scores. For a more detailed overview of the heuristic evaluation refer to this report: https://github.com/ArtemPoliakov51/OTP1/blob/main/Documents/product_documents/attendance_checker_heuristic_evalutation.pdf

What Went Well

All tasks that were planned were completed on time and nothing was postponed to the next and final sprint. The software was accepted with User Acceptance Testing and heuristic evaluation. It meets all the requirements and criteria that were set at the beginning of the

project. No major bugs or issues were encountered. All the team members participated in UAT and heuristic evaluation.

What Could Be Improved

Team communication has been an issue throughout the whole project and no improvement was seen in this sprint. Task distribution and participation was quite unbalanced between the members.

The documentation could have been better for this sprint. The codebase is also still quite messy and has a lot of maintainability and best practices issues.

Work Hours

Team member	Assigned Tasks	Time Spent (hrs)	In-class assignments
Riikka Kautonen	Code clean up (duplicate lines) Test case creation, UAT + report, Heuristic evaluation + report, SonarQube Jenkins pipeline, Project management, Sprint 7 review report	19	Submitted
Artem Poliakov	Code clean up, testing, Heuristic evaluation & report.	10~	Not submitted
Jere Pyörökivi	Code cleanup (labels, duplicate lines), UAT, Heuristic evaluation & report	12	Submitted
Zachris Zweygberg	User acceptance testing, heuristic evaluation & report, code clean up	10	Submitted